

# Introduction To Networking and The OSI Model

1. Draw an OSI Model diagram showing all 7 layers, and for each layer, write one real-world example (such as HTTP for Application, TCP for Transport, Ethernet for Data Link, cables for Physical, etc).

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## OSI Model (7 Layers)

The OSI (Open Systems Interconnection) model is a 7-layer framework that standardizes how data is transmitted between devices over a network.

| Layer                                     | Function                                                                                                     | Real-World Example                                                                                                                   |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>7</b> <b>Application</b><br>(Layer 7)  | Provides network services to end-user applications. Handles data format, encryption, and session management. | <b>HTTP / HTTPS</b><br>(web browsing)             |
| <b>6</b> <b>Presentation</b><br>(Layer 6) | Handles data translation, compression and encryption. (ensures data is in a usable format).                  | <b>SSL / TLS</b><br>(encryption)                 |
| <b>5</b> <b>Session</b><br>(Layer 5)      | Manages session establishment, maintenance and termination. (between applications).                          | <b>NetBIOS / RPC</b><br>(session management)    |
| <b>4</b> <b>Transport</b><br>(Layer 4)    | Ensures reliable or best-effort data delivery, handles flow control, error recovery and segmentation.        | <b>TCP / UDP</b><br>(reliable / fast delivery)  |
| <b>3</b> <b>Network</b><br>(Layer 3)      | Handles logical addressing, routing and path selection across networks.                                      | <b>IP (IPv4 / IPv6)</b><br>(routing)            |
| <b>2</b> <b>Data Link</b><br>(Layer 2)    | Provides framing, MAC addressing, error detection and access to the physical medium.                         | <b>Ethernet (802.3)</b><br>(LAN communication)  |
| <b>1</b> <b>Physical</b><br>(Layer 1)     | Defines the physical medium, signal, voltage, and data rate for transmission.                                | <b>Network Cable (Cat6)</b><br>(copper cable)   |

### How Data Flows Through the OSI Layers



2. List three apps you use daily (for example: Instagram, Zomato, or WhatsApp) and for each, pick one network activity (like sending a message or placing an order) and explain which OSI layers are involved in that activity.

->Here is a simple answer you can use:

| App       | Network Activity     | OSI Layers Involved                                                                                                                                                    |
|-----------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WhatsApp  | Sending a message    | <b>Application:</b> WhatsApp messaging service; <b>Transport:</b> TCP/UDP; <b>Network:</b> IP; <b>Data Link:</b> Wi-Fi/Ethernet; <b>Physical:</b> radio signals/cables |
| Instagram | Loading a photo      | <b>Application:</b> HTTP/HTTPS; <b>Transport:</b> TCP/QUIC; <b>Network:</b> IP; <b>Data Link:</b> Wi-Fi/Ethernet; <b>Physical:</b> radio signals/cables                |
| Zomato    | Placing a food order | <b>Application:</b> HTTPS; <b>Transport:</b> TCP/QUIC; <b>Network:</b> IP; <b>Data Link:</b> Wi-Fi/Ethernet; <b>Physical:</b> radio signals/cables                     |

#### Explanation

All seven OSI layers can participate in network communication. **Application, Transport, and Network** layers handle services, data delivery, and addressing, while **Data Link and Physical** layers handle communication across the local network and physical medium. The **Presentation and Session** layers describe functions such as data formatting/encryption and managing communication sessions, although modern apps often implement these functions through protocols and software that don't map neatly to separate OSI layers.

3. Install and open Wireshark on your computer, start a capture, then visit any website (for example, [www.example.com](http://www.example.com)) in your browser. Stop the capture and save the PCAP file

#### ->Wireshark Website Traffic Capture

##### 1. Install Wireshark

- Download and install Wireshark from the official Wireshark website.
- During installation, allow the required packet-capture driver (Npcap) to be installed.

##### 2. Start Wireshark

- Open Wireshark.

- Select your active network interface, such as **Wi-Fi** or **Ethernet**.

### 3. Start the capture

- Click the blue **Start** button.
- Open your web browser in a separate window.

### 4. Visit a website

- Go to **example.com**.
- Wait for the webpage to load completely.
- Return to Wireshark after a few seconds.

### 5. Stop the capture

- Click the red **Stop** button in Wireshark.

### 6. Save the PCAP file

- Select **File → Save As**.
- Give the file a name such as:  
website\_capture.pcapng
- Save it somewhere easy to find, such as your Documents folder.

## Documentation

Record:

| Item              | Result                 |
|-------------------|------------------------|
| Network Interface | _____                  |
| Website Visited   | example.com            |
| Capture Started   | _____                  |
| Capture Stopped   | _____                  |
| File Name         | website_capture.pcapng |
| File Format       | PCAPNG                 |

 Take screenshots of:

- Wireshark while capturing packets.
- The completed packet list after stopping the capture.
- The saved .pcapng file in your folder.

**Note:** A .pcapng file contains network traffic from your capture, so don't share it publicly if you're capturing on a personal or organizational network.

4. In Wireshark, find a captured HTTP request packet from your browsing session, take a screenshot of the packet details, and annotate which parts of the packet correspond to the Application, Transport, Network, and Data Link layers.  
  
**Hint:** Use the packet details pane in Wireshark to expand each protocol section (HTTP, TCP, IP, Ethernet).

#### ->Wireshark HTTP Packet Layer Identification

Use the packet you captured in the previous Wireshark exercise.

##### 1. Open the capture

- Open your .pcapng file in Wireshark.
- In the display-filter bar, enter:  
http
- Press **Enter**.

##### 2. Select an HTTP request

- Look for a packet such as **GET** or **POST**.
- Click the packet to display its details.

##### 3. Expand the Packet Details pane

Identify these sections:

| OSI Layer             | Wireshark Protocol | What to Identify                          |
|-----------------------|--------------------|-------------------------------------------|
| Application – Layer 7 | HTTP               | GET/POST request, Host, URI, HTTP headers |

| OSI Layer           | Wireshark Protocol | What to Identify                                          |
|---------------------|--------------------|-----------------------------------------------------------|
| Transport – Layer 4 | TCP                | Source/destination ports, sequence number, acknowledgment |
| Network – Layer 3   | IPv4/IPv6          | Source and destination IP addresses                       |
| Data Link – Layer 2 | Ethernet II        | Source and destination MAC addresses, frame information   |

### Example Annotation

Ethernet II → Data Link (Layer 2)



IPv4 → Network (Layer 3)



TCP → Transport (Layer 4)



HTTP → Application (Layer 7)

 **Screenshot requirement:** Take a screenshot with the selected HTTP packet and the **Packet Details** pane visible. Expand **Ethernet II**, **Internet Protocol**, **TCP**, and **HTTP**, then annotate each section with its corresponding OSI layer.

**Important:** If you visited an HTTPS website, Wireshark may show **TLS** rather than readable HTTP. In that case, use an HTTP capture from a site/session that actually contains HTTP traffic; don't expect the HTTP contents to be visible inside encrypted HTTPS traffic.